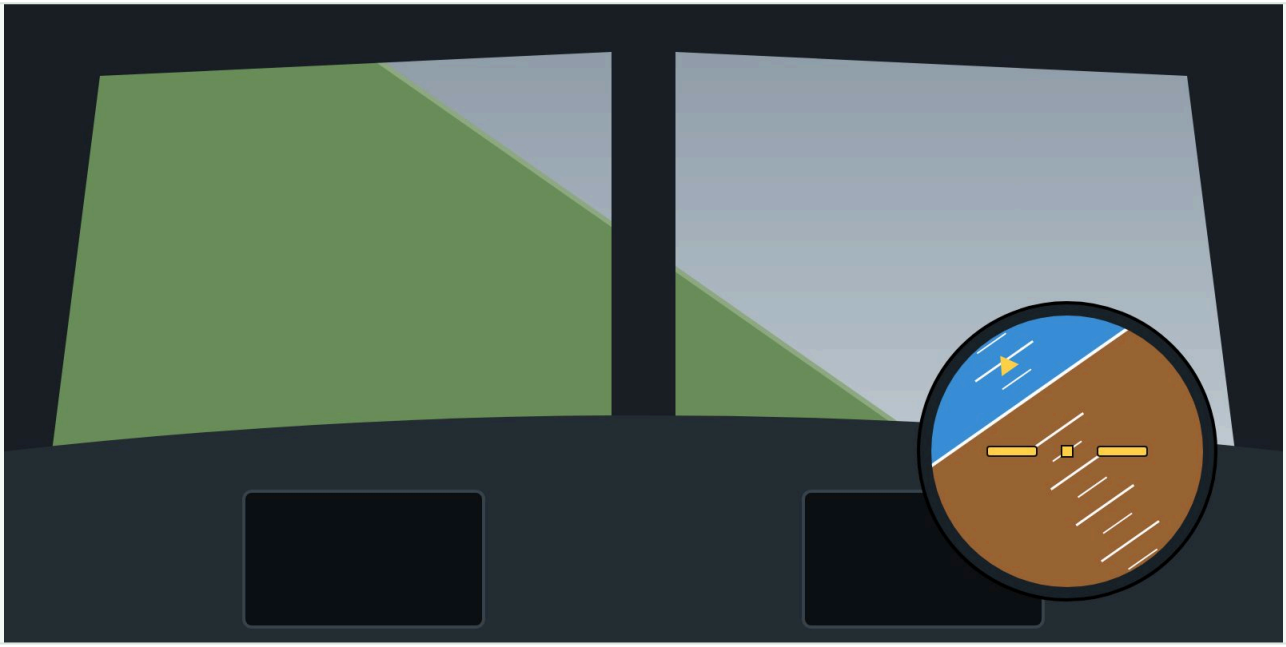
MODULE 5 · PREVENTION

# Lesson 17: Recognising a Developing Upset

Lesson 17 of 24 · about 8 min

**LEARNING OBJECTIVES**

- Identify the flight-deck cues of a developing upset.
- Read an attitude indicator quickly and correctly in an unusual attitude.
- Confirm the situation by cross-checking instruments.

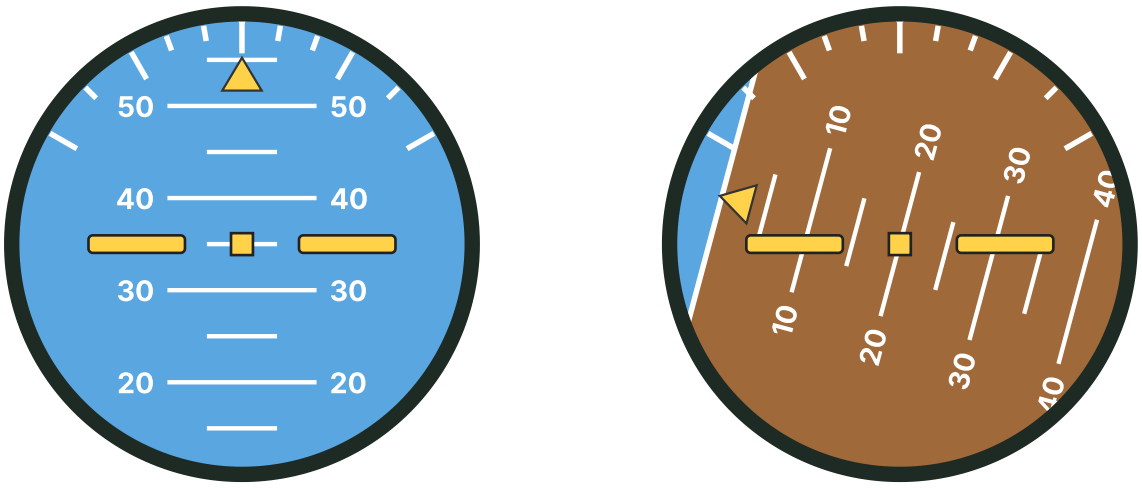


Recognise early — attitude first, then confirm.

## Cues of a developing upset

- **Speed:** trend vector towards the low-speed band or the overspeed limit; speed not stabilising after a thrust change.
- **Attitude:** pitch or bank increasing without an intended input; pitch limit indicator approaching the aircraft symbol.
- **Energy:** thrust at idle when it should be climbing; pitch high and speed decaying.
- **Automation:** unexpected mode changes, autopilot or autothrust disconnect, flight control law changes.
- **Physical:** buffet, unusual noise, unexpected control forces, g changes.
- **Warnings:** stall warning, overspeed, bank angle aural alert, windshear alert.

## Reading the attitude indicator




**Nose-high, wings level** **Nose-low, high bank**

Recognise the upset from the attitude indicator first. Left: about 35° nose-up. Right: about 20° nose-down with about 75° of right bank — the sky pointer points to the sky, so roll towards it.

- **Look at the attitude indicator first.** Determine pitch (sky or ground? which pitch line?) and bank (where is the sky pointer?).
- The **sky pointer** (or bank pointer) always points towards the sky. In a nose-low, high-bank upset, **roll towards the sky pointer** to bring the wings level.
- At extreme attitudes, the display may show mostly sky or mostly ground, and chevrons may point the way back to the horizon. Know your PFD.

## Confirm before you act

- Cross-check both primary attitude indicators and the **standby** instrument. A single failed indicator can create a false upset.
- Confirm the trend with airspeed, altitude and vertical speed. A nose-high attitude with decaying speed and a climbing altimeter is very different from a nose-high attitude with a rapid descent (possible stall).
- Say what you see: “Nose high, wings level, speed decreasing.” This helps both pilots build the same picture.

> **ACT EARLY**
> If you recognise a developing upset, correct it immediately with small, smooth inputs. A small correction now prevents a large recovery later.

**KEY TAKEAWAYS**

- Developing upsets show up as speed, attitude, energy, automation, physical and warning cues.
- Read the attitude indicator first; the sky pointer always points to the sky.
- Confirm with the other attitude indicator, the standby instrument and the speed, altitude and vertical speed trends.

**KNOWLEDGE CHECK**

**In a nose-low, high-bank upset, which way should you roll to bring the wings level?**

A. Away from the sky pointer

B. Towards the sky pointer — the shortest direction to wings level

C. Always to the left

D. It does not matter

**Answer: B.** The sky pointer points towards the sky. Rolling towards it brings the lift vector up and the wings level by the shortest way.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.